



Chapter 19

Camada de Rede: Endereçamento Lógico

19-1 IPv4 ADDRESSES

*Um endereço **IPv4** é um endereço de **32 bits** que define de forma exclusiva e universal a conexão de um dispositivo (por exemplo, um computador ou um roteador) para a Internet.*

Topics discussed in this section:

Espaço de Endereçamento

Notações

Endereçamento com classes (Classful Addressing)

Endereçamento sem classes (Classless Addressing)

Network Address Translation (NAT)



Note

Um endereço IPv4 tem 32 bits de comprimento.



Note

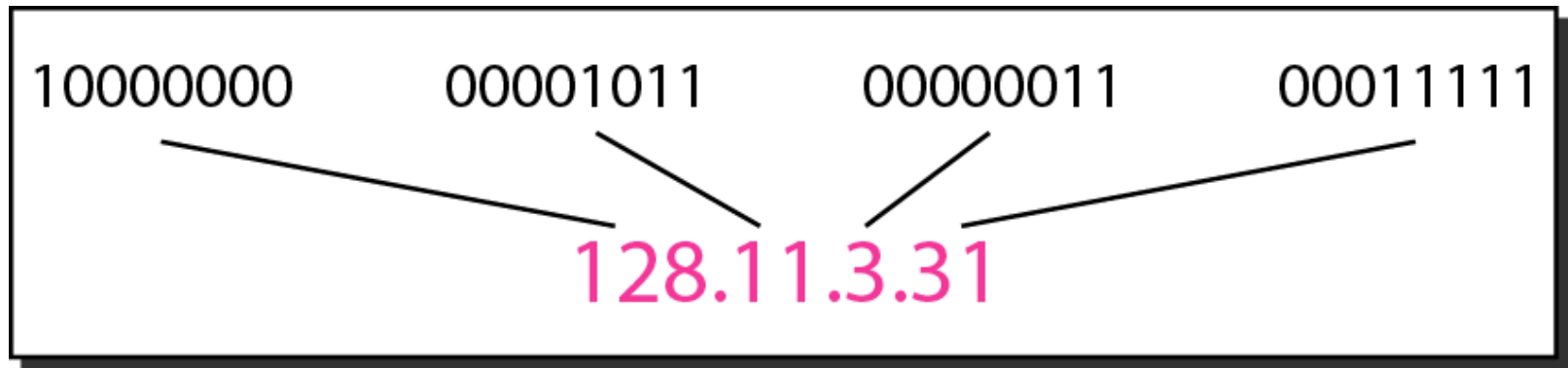
Os endereços IPv4 são únicos e exclusivos.



Note

**Os espaço de endereços do IPv4 é
 2^{32} ou 4,294,967,296.**

Figure 19.1 *Dotted-decimal notation and binary notation for an IPv4 address*





Example 19.1

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution

We replace each group of 8 bits with its equivalent decimal number (see Appendix B) and add dots for separation.

a. 129.11.11.239

b. 193.131.27.255



Example 19.2

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

Solution

We replace each decimal number with its binary equivalent (see Appendix B).

a. 01101111 00111000 00101101 01001110

b. 11011101 00100010 00000111 01010010



Example 19.3

Find the error, if any, in the following IPv4 addresses.

- a. 111.56.045.78
- b. 221.34.7.8.20
- c. 75.45.301.14
- d. 11100010.23.14.67

Solution

- a. There must be no leading zero (045).***
- b. There can be no more than four numbers.***
- c. Each number needs to be less than or equal to 255.***
- d. A mixture of binary notation and dotted-decimal notation is not allowed.***

Note

**No endereçamento com classes, o espaço de endereços é dividido em cinco classes :
A, B, C, D, and E.**

Figure 19.2 *Finding the classes in binary and dotted-decimal notation*

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0-127			
Class B	128-191			
Class C	192-223			
Class D	224-239			
Class E	240-255			

b. Dotted-decimal notation



Example 19.4

Find the class of each address.

- a.* 00000001 00001011 00001011 11101111
- b.* 11000001 10000011 00011011 11111111
- c.* 14.23.120.8
- d.* 252.5.15.111

Solution

- a.* The first bit is 0. This is a class A address.
- b.* The first 2 bits are 1; the third bit is 0. This is a class C address.
- c.* The first byte is 14; the class is A.
- d.* The first byte is 252; the class is E.

Table 19.1 *Number of blocks and block size in classful IPv4 addressing*

<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

Note

No endereçamento com classes, uma grande parte dos endereços disponíveis foram desperdiçados.

Table 19.2 *Default masks for classful addressing*

<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>	<i>CIDR</i>
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24



Note

O endereçamento com classes, que se tornou quase obsoleto, foi substituído por endereçamento sem classes

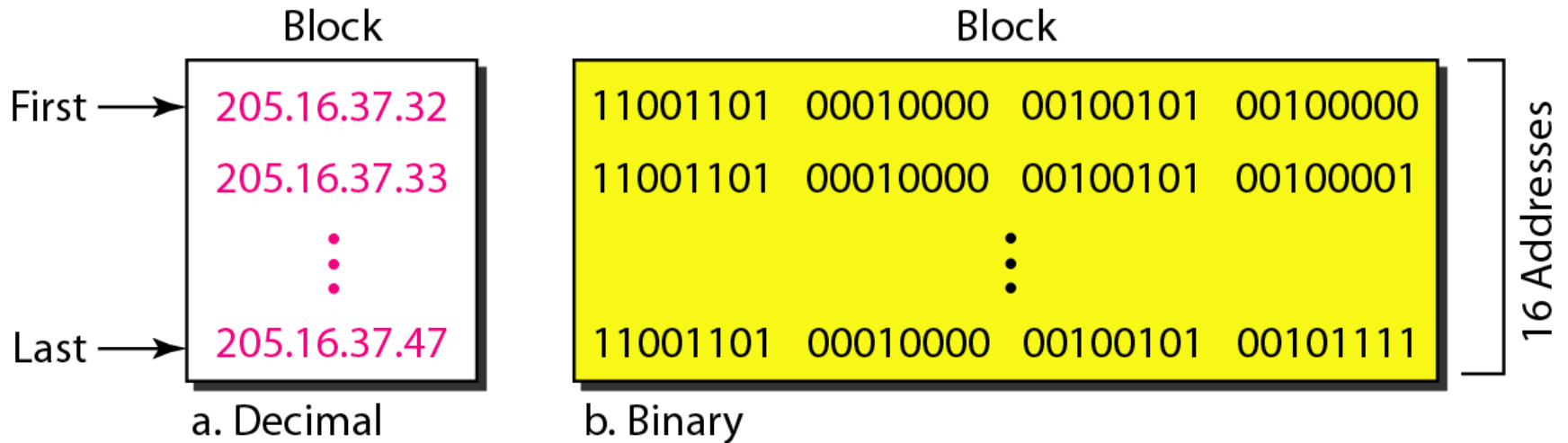


Example 19.5

Figure 19.3 shows a block of addresses, in both binary and dotted-decimal notation, granted to a small business that needs 16 addresses.

We can see that the restrictions are applied to this block. The addresses are contiguous. The number of addresses is a power of 2 ($16 = 2^4$), and the first address is divisible by 16. The first address, when converted to a decimal number, is 3,440,387,360, which when divided by 16 results in 215,024,210.

Figure 19.3 *A block of 16 addresses granted to a small organization*



Note

No endereçamento IPv4, um bloco de "endereço" pode ser definido como
x.y.z.t/n
em que x.y.z.t define um dos endereços
e o /n define a máscara.

Note

O primeiro endereço no bloco pode ser encontrado configurando os bits 32 - n mais à direita para 0s.



Example 19.6

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32–28 rightmost bits to 0, we get

11001101 00010000 00100101 00100000

or

205.16.37.32.

This is actually the block shown in Figure 19.3.

Note

O último endereço no bloco pode ser encontrado configurando os bits 32 - n mais à direita para 1s.



Example 19.7

Find the last address for the block in Example 19.6.

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32 – 28 rightmost bits to 1, we get

11001101 00010000 00100101 00101111

or

205.16.37.47

This is actually the block shown in Figure 19.3.



Note

O número de endereços no bloco pode ser encontrado usando a fórmula

$$2^{32-n}.$$



Example 19.8

Find the number of addresses in Example 19.6.

Solution

The value of n is 28, which means that number of addresses is 2^{32-28} or 16.



Example 19.9

Another way to find the first address, the last address, and the number of addresses is to represent the mask as a 32-bit binary (or 8-digit hexadecimal) number. This is particularly useful when we are writing a program to find these pieces of information. In Example 19.5 the /28 can be represented as

11111111 11111111 11111111 11110000

(twenty-eight 1s and four 0s).

Find

- a.** *The first address*
- b.** *The last address*
- c.** *The number of addresses.*

Example 19.9 (continued)

Solution

- a. The first address can be found by ANDing the given addresses with the mask. ANDing here is done bit by bit. The result of ANDing 2 bits is 1 if both bits are 1s; the result is 0 otherwise.*

Address:	11001101	00010000	00100101	00100111
Mask:	11111111	11111111	11111111	11110000
First address:	11001101	00010000	00100101	00100000

Example 19.9 (continued)

- b.** *The last address can be found by ORing the given addresses with the complement of the mask. ORing here is done bit by bit. The result of ORing 2 bits is 0 if both bits are 0s; the result is 1 otherwise. The complement of a number is found by changing each 1 to 0 and each 0 to 1.*

Address:	11001101	00010000	00100101	00100111
Mask complement:	00000000	00000000	00000000	00001111
Last address:	11001101	00010000	00100101	00101111

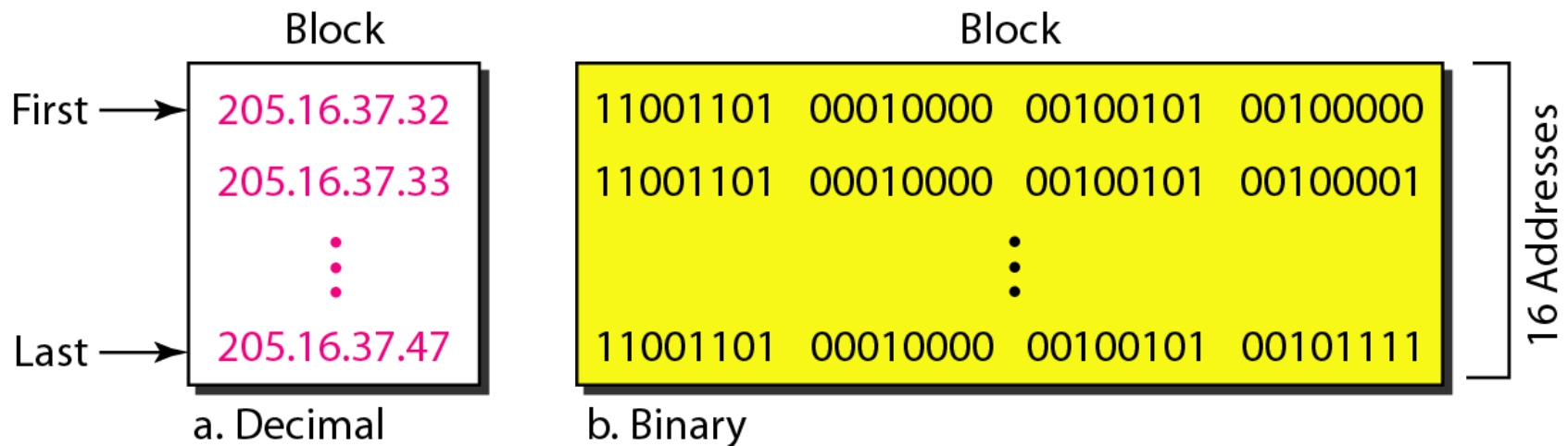


Example 19.9 (continued)

- c. The number of addresses can be found by complementing the mask, interpreting it as a decimal number, and adding 1 to it.*

Mask complement:	00000000	00000000	00000000	00001111
Number of addresses:	$15 + 1 = 16$			

Figure 19.4 *A network configuration for the block 205.16.37.32/28*



Note

O primeiro endereço em um bloco normalmente não é atribuído a nenhum dispositivo; ele é usado como o endereço de rede que representa a organização para o resto do mundo

Figure 19.4 *Uma configuração de rede para o bloco 205.16.37.32/28*

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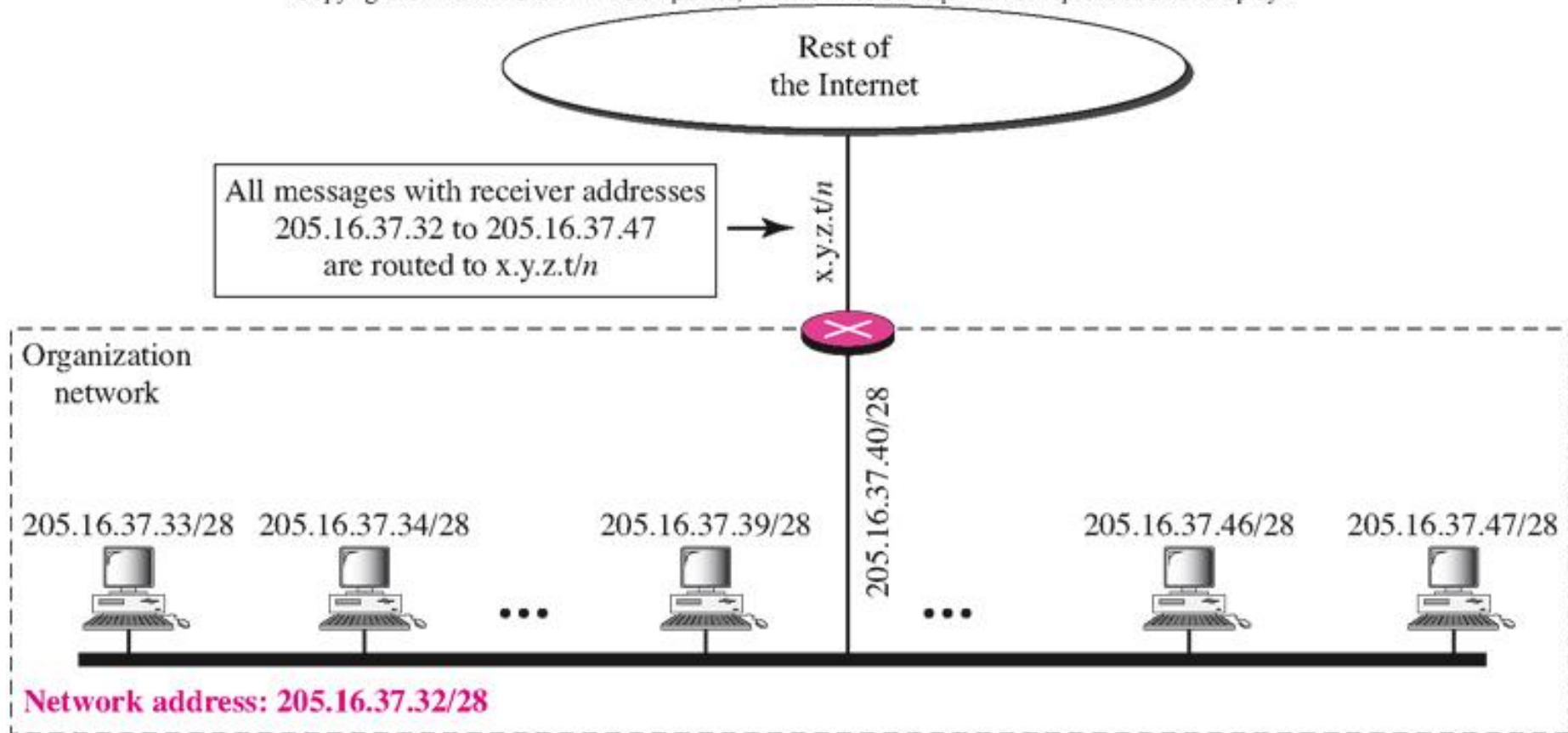


Figure 19.5 *Two levels of hierarchy in an IPv4 address*

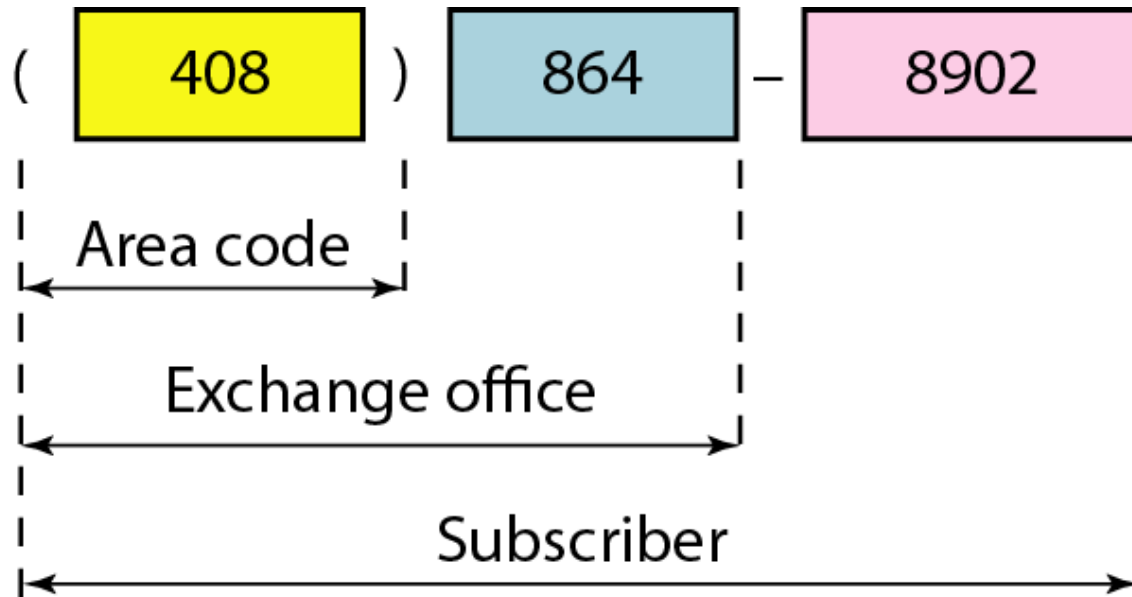
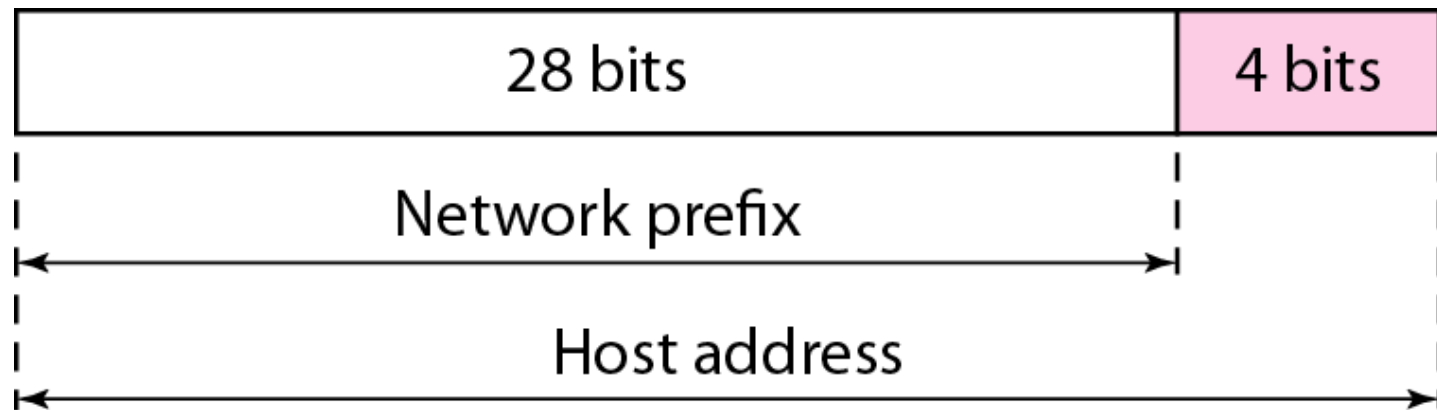


Figure 19.6 *A frame in a character-oriented protocol*



Note

Cada endereço no bloco pode ser considerado como uma estrutura hierárquica de dois níveis: os n bits mais à esquerda (prefixo) definem a rede; os bits $32 - n$ mais à direita definem o host

Figure 19.7 *Configuration and addresses in a subnetted network*

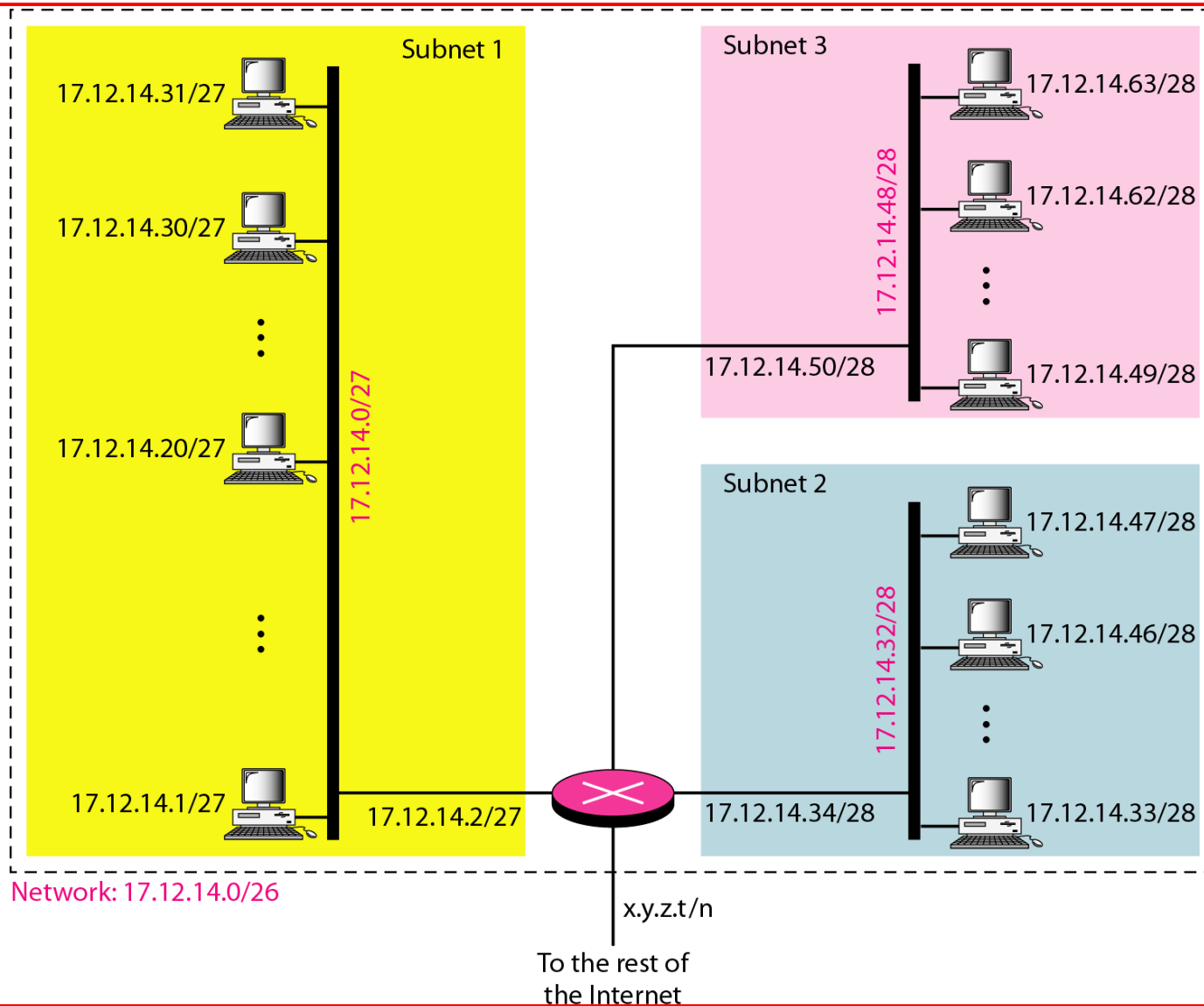
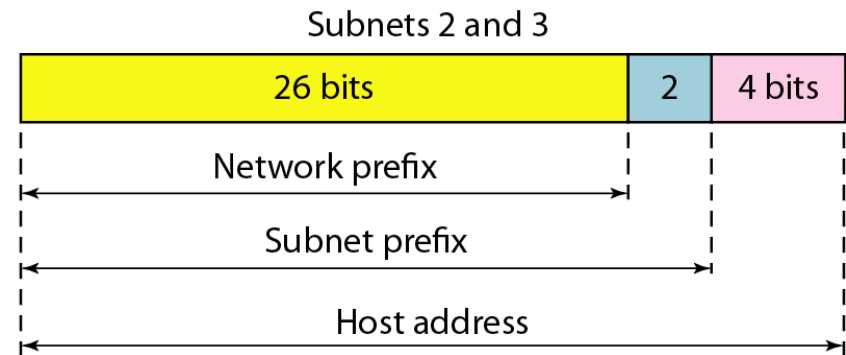
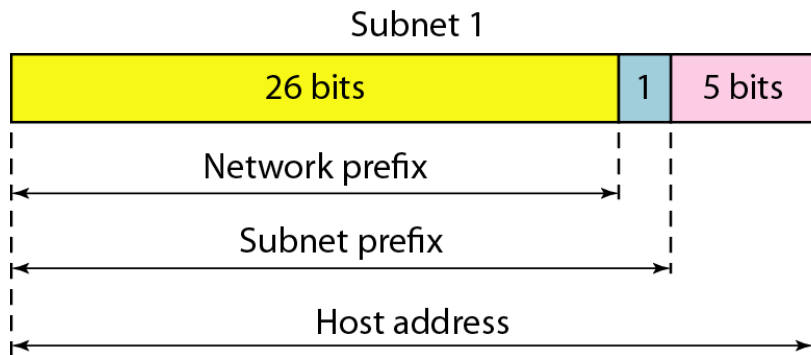


Figure 19.8 *Three-level hierarchy in an IPv4 address*





Example 19.10

Um ISP recebe um bloco de endereços começando com 190.100.0.0/16 (65.536 endereços). O ISP precisa distribuir esses endereços para três grupos de clientes da seguinte forma:

- a.** O primeiro grupo tem 64 clientes; cada um precisa de 256 endereços.*
- b.** O segundo grupo tem 128 clientes; cada um precisa de 128 endereços.*
- c.** O terceiro grupo tem 128 clientes; cada um precisa de 64 endereços.*

Defina os subblocos e descubra quantos endereços ainda estão disponíveis após essas alocações.



Example 19.10 (continued)

Solution

Figure 19.9 shows the situation.

Group 1

Para este grupo, cada cliente precisa de 256 endereços. Isso significa que 8 bits são necessários para definir cada host. O comprimento do prefixo é então de $32 - 8 = 24$. Os endereços são:

<i>1st Customer:</i>	<i>190.100.0.0/24</i>	<i>190.100.0.255/24</i>
<i>2nd Customer:</i>	<i>190.100.1.0/24</i>	<i>190.100.1.255/24</i>
<i>...</i>		
<i>64th Customer:</i>	<i>190.100.63.0/24</i>	<i>190.100.63.255/24</i>
<i>Total = $64 \times 256 = 16,384$</i>		



Example 19.10 (continued)

Group 2

Para este grupo, cada cliente precisa de 128 endereços. Isso significa que 7 bits são necessários para definir cada host. O comprimento do prefixo é então de $32 - 7 = 25$. Os endereços são

<i>1st Customer:</i>	<i>190.100.64.0/25</i>	<i>190.100.64.127/25</i>
<i>2nd Customer:</i>	<i>190.100.64.128/25</i>	<i>190.100.64.255/25</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.127.128/25</i>	<i>190.100.127.255/25</i>
<i>Total = $128 \times 128 = 16,384$</i>		

Example 19.10 (continued)

Group 3

Para este grupo, cada cliente precisa de 64 endereços. Isso significa que 6 bits ($\log_2 64$) são necessários para cada host. O comprimento do prefixo é então de $32 - 6 = 26$. Os endereços são

<i>1st Customer:</i>	<i>190.100.128.0/26</i>	<i>190.100.128.63/26</i>
<i>2nd Customer:</i>	<i>190.100.128.64/26</i>	<i>190.100.128.127/26</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.159.192/26</i>	<i>190.100.159.255/26</i>
<i>Total = $128 \times 64 = 8192$</i>		

Número de endereços concedidos ao ISP: 65.536

Número de endereços alocados pelo ISP: 40.960

Número de endereços disponíveis: 24.576

Figure 19.9 *An example of address allocation and distribution by an ISP*

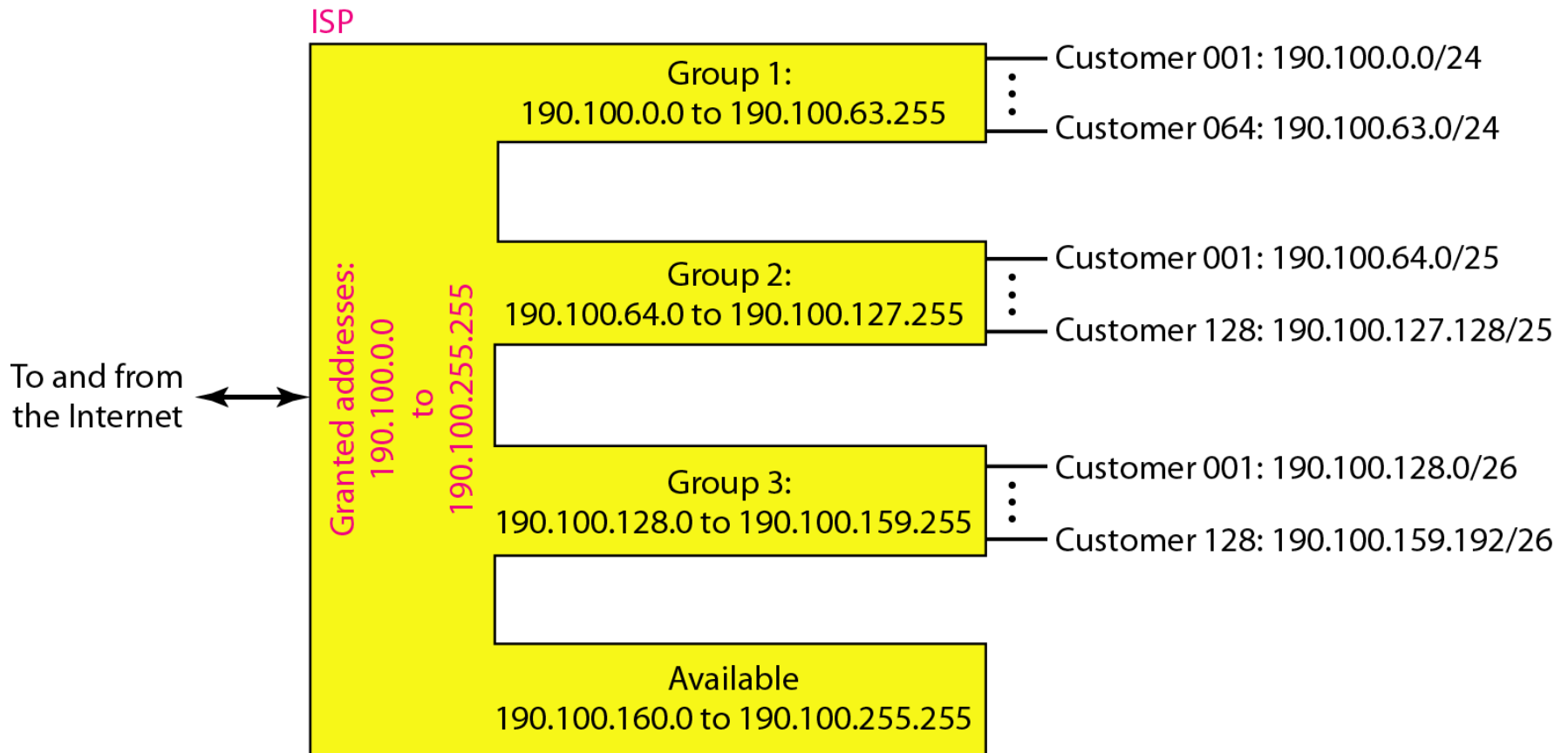


Table 19.3 *Addresses for private networks*

<i>Range</i>			<i>Total</i>
10.0.0.0	to	10.255.255.255	2^{24}
172.16.0.0	to	172.31.255.255	2^{20}
192.168.0.0	to	192.168.255.255	2^{16}

Figure 19.10 *A NAT implementation*

Site using private addresses

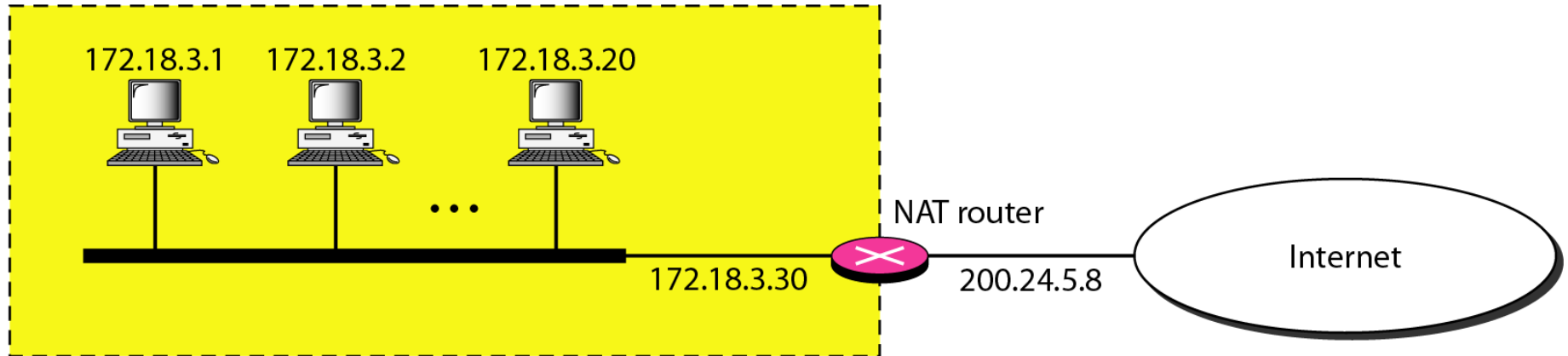


Figure 19.11 *Addresses in a NAT*

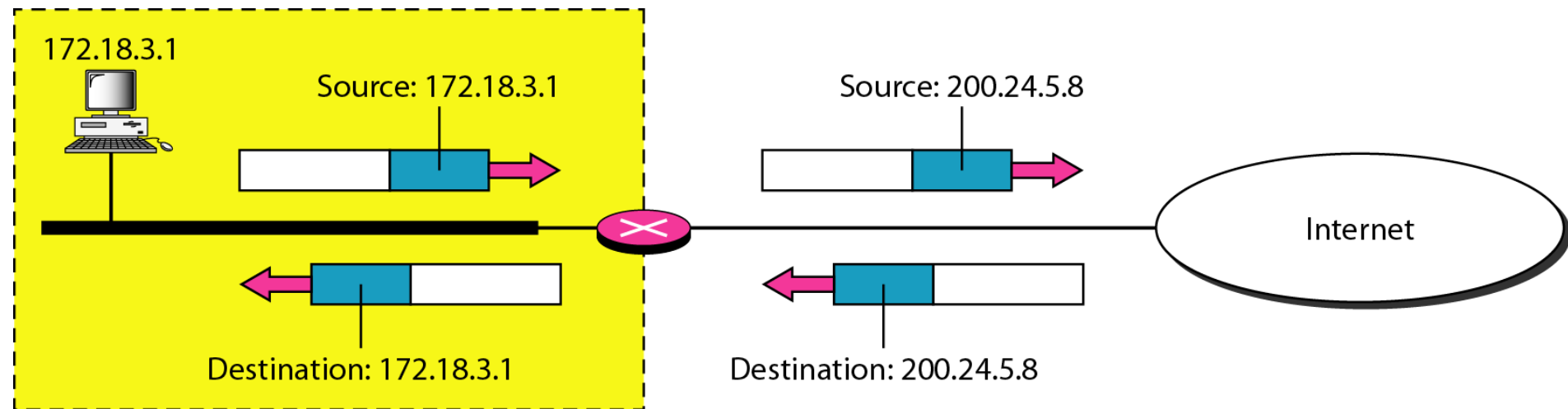


Figure 19.12 NAT address translation

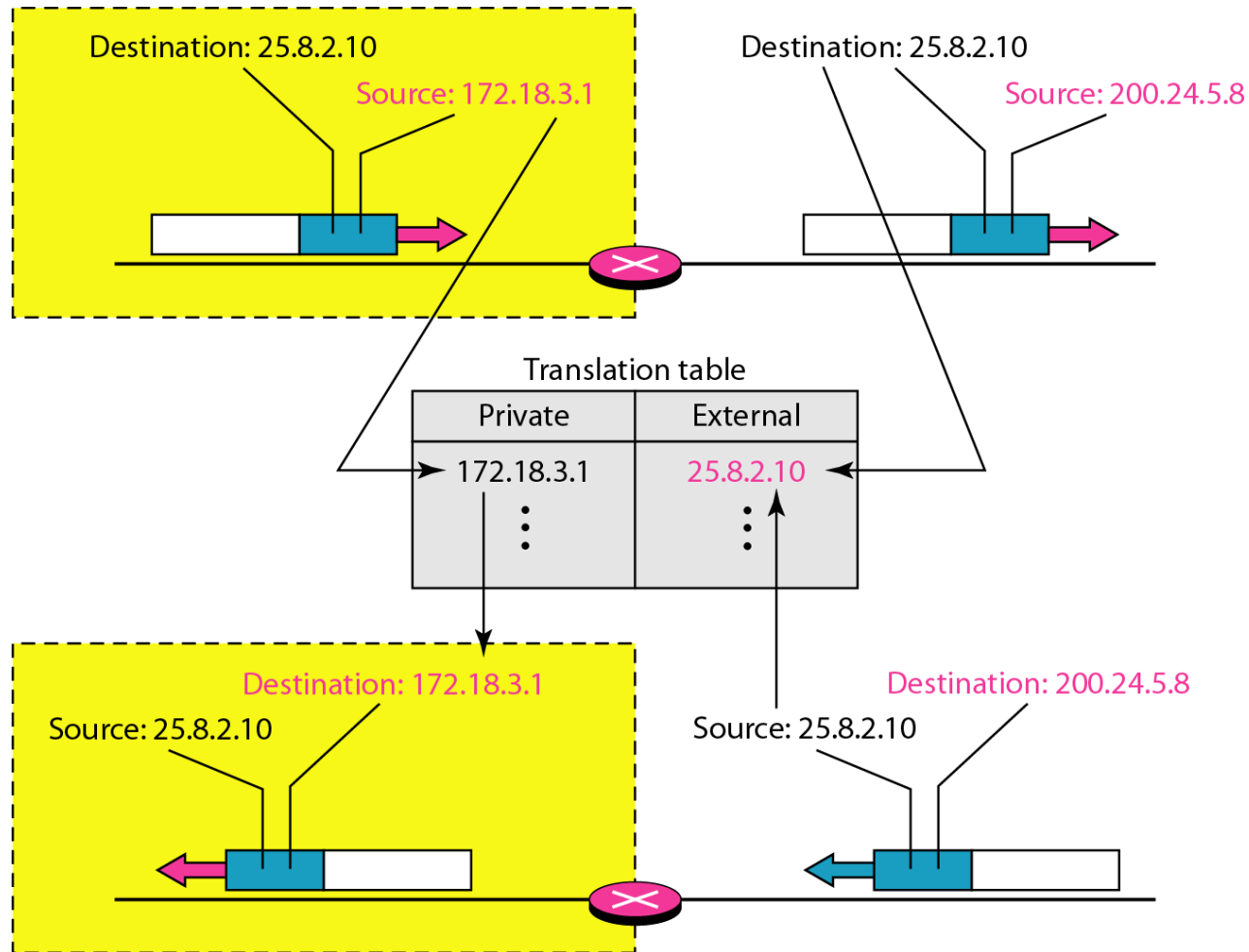


Table 19.4 *Five-column translation table*

<i>Private Address</i>	<i>Private Port</i>	<i>External Address</i>	<i>External Port</i>	<i>Transport Protocol</i>
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
...	...	...	...	...

Figure 19.13 *An ISP and NAT*

